

6.5.1.3

EE24BTECH11013 - MANIKANTA D

Question:

Find the Maximum and Minimum values of the function if exists

$$y(x) = -(x - 1)^2 + 10$$

Solution :

Theoretical Solution :

Finding the derivative of the function

$$\frac{dy}{dx} = -2(x - 1) \quad (0.1)$$

Setting $\frac{dy}{dx} = 0$, we get:

$$x = 1 \quad (0.2)$$

At $x = 1$, substituting into $y(x)$:

$$y(1) = -(1 - 1)^2 + 10 = 10 \quad (0.3)$$

To determine the nature of this critical point, we find the second derivative:

$$\frac{d^2y}{dx^2} = -2 \quad (0.4)$$

Since $\frac{d^2y}{dx^2} < 0$, the function has a maximum at $x = 1$.

The function is a downward-opening parabola, so there is no absolute minimum. The function decreases without bound as $x \rightarrow \pm\infty$.

Computational Solution :

Maximum value of the function can be computed using the **Gradient ascent** method:

Update the position iteratively:

$$x_{n+1} = x_n + \eta \cdot \frac{dy}{dx} \quad (0.5)$$

Taking $x_0 = 0$ as the starting point

Here, $\eta = 0.1$, where η is the learning rate.

$$x_{n+1} = x_n + \eta \cdot (-2(x_n - 1)) \quad (0.6)$$

The gradient ascent method converges to $x \approx 1$, and the maximum value is:

$$y(1) \approx 10 \quad (0.7)$$

Minimum value of the function cannot be computed because the function decreases indefinitely as $x \rightarrow \pm\infty$.

Computational Results :

- Absolute Maximum:

$$x \approx 1, y(x) \approx 10 \quad (0.8)$$

